

C++2: Lesson 8 – Lab Project

1. Add **summary** at the top, **documentation** at the bottom and **comments** where necessary.
- Recordings and interviews are an exception, but they should be included in your submission code.
2. **Cite** the source of your information.
3. Refrain from including anything **you don't fully understand**.

2

200 points

Write a program that simulates a **web browser's navigation system** using both a **Stack (LIFO)** and a **Queue (FIFO)**. The program should allow users to:

1. **Visit a new webpage** – Adds the current page to the **back history** stack and clears the forward queue.
2. **Go back** – Moves the last visited page from the **back stack** to the **forward queue** (LIFO behavior).
3. **Go forward** – Restores the earliest removed page from the **forward queue** (FIFO behavior).
4. **Exit** – Ends the program.

Expected Program Flow

1. The program **displays a menu** with navigation options.
2. **The user visits new webpages**, adding them to browsing history.
3. The user **navigates back or forward** through previously visited pages
4. The program continues until the user chooses to exit.

Menu:

1. Visit a new webpage
2. Go back
3. Go forward
4. Exit

Enter choice: 1

Enter a URL to visit: google.com

Enter choice: 1

Enter a URL to visit: youtube.com

Enter choice: 1

Enter a URL to visit: github.com

Enter choice: 2

Going back to: youtube.com

Enter choice: 2

Going back to: google.com

Enter choice: 3

Going forward to: youtube.com

Enter choice: 4

Exiting browser...

File(s) to
submit

C208_2.h
C208_2.cpp
C208_2.png